

# HARINARAYANAN U

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## Summary

Software engineer with production experience building AI-powered applications, backend services, and distributed systems. Experienced in developing RAG and LLM-powered pipelines, designing reliable REST APIs, evaluating AI systems, and deploying production services using Python, Docker, and AWS. Strong builder mindset with a passion for turning ideas into working software and solving complex engineering problems.

## Education

### Amrita Vishwa Vidyapeetham

Nov. 2022 – Aug. 2026

*Bachelor of Technology in Computer Science and Engineering | CGPA: 8.28/10*

*Kollam, Kerala, India*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design (C++/Java), Operating Systems, Database Management Systems, Computer Networks, Distributed Systems, Design and Analysis of Algorithms

## Experience

### RevGain AI

Sept. 2025 – Feb. 2026

*Software Developer Intern*

*Hyderabad, India*

- Designed and shipped a scalable AI-powered Text-to-SQL service using FastAPI and PostgreSQL, deployed with Docker on AWS EC2 for multi-tenant enterprise workloads.
- Engineered a reliable REST API (<600 ms p95 latency) with idempotent request handling, schema-aware SQL generation, Pydantic validation, and read-only database access controls.
- Improved SQL generation accuracy to ~85% on complex multi-join queries through schema-linking, few-shot prompting, and retrieval-augmented generation, reducing error rates by ~40% versus a zero-shot baseline through iterative evaluation.
- Collaborated with product and data teams to gather requirements, evaluate AI outputs, validate integrations through structured testing, and deliver production features in Agile development cycles.

### TiVo

Aug. 2024 – Nov. 2024

*Software Developer Intern*

*Bangalore, India*

- Diagnosed and resolved live service buffering issues in AVPlayer by refactoring caching, prefetch logic, and player state handling in Swift, reducing buffering incidents by ~25% at scale.
- Contributed to 50+ collaborative code reviews and delivered 8+ SwiftUI component redesigns, improving code quality, maintainability, and engineering practices.

## Projects

### QueueCtl | Python, SQLite WAL, Flask, Socket.IO, Docker | GitHub

Nov. 2025 – Present

- Designed and published a broker-free distributed job queue on PyPI using atomic job claiming to ensure exactly-once task execution across concurrent workers.
- Implemented exponential backoff retries, Dead Letter Queues (DLQ), DAG-based dependency resolution, and cron scheduling for reliable asynchronous workflow execution.
- Built HMAC-signed webhooks and a real-time observability dashboard to securely deliver events and monitor queue health, worker activity, latency, and execution metrics.

### SONAR | FastAPI, PyTorch, Transformers, React, Docker

Mar. 2026

- Designed and implemented a multimodal AI platform integrating EmoRoBERTa-X, DeBERTa, and WavLM with confidence-aware fusion for personalized music recommendation.
- Built a production FastAPI inference pipeline with SHAP-based explainability for scalable transformer-based model serving.

## Publications

### SONAR: Hybrid Explainable Emotion-Aware Music Recommendation | IEEE Xplore

Mar. 2026

- IEEE IATMSI 2026 — Published work on explainable multimodal emotion-aware recommendation using transformer-based text/audio models and confidence-aware fusion.

### PrivMod: Federated Learning for Privacy-Preserving Content Moderation | IEEE Xplore

Dec. 2025

- IEEE I4Tech 2025 — Developed a privacy-preserving federated learning framework achieving 94% accuracy across 100K+ samples using a CNN-BERT moderation pipeline.

## Technical Skills

**Languages:** Python, Java, Go, C/C++, JavaScript, SQL

**AI/ML:** PyTorch, Transformers, LLM Integration, Retrieval-Augmented Generation (RAG)

**Backend & Systems:** FastAPI, REST APIs, Django REST, Microservices, Distributed Systems, PostgreSQL, SQLite, Fault Tolerance

**Tools & Cloud:** Docker, AWS EC2, Git, GitHub Actions, Linux, CI/CD

**Concepts:** Object-Oriented Design, Data Structures & Algorithms, Operating Systems, Computer Networks, Concurrency